

Cinema Database: Data-Related Operations Document

1. Movie Rating Consistency Check (Detect Suspicious Ratings)

Description: Detects cases where a customer submits a review for a movie but does not have a valid ticket for any past session of that movie. This identifies invalid, fraudulent, or mistaken reviews that should be removed or flagged.

Purpose: Ensures the review system remains trustworthy by catching reviews that violate business rules (“customer must have attended the movie to review it”). Helps admins maintain data quality.

System Requirements: Compare reviews with session and ticket history to confirm valid attendance; identify mismatches.

2. Browse Movies and View Details

Description: Displays all movies with genre, duration, description, and aggregated rating based on reviews.

Purpose: Supports customer browsing and movie discovery.

System Requirements: Join movies with reviews and calculate average rating dynamically.

3. View Upcoming Sessions for a Selected Movie

Description: Lists all showtimes (sessions) for a movie, including hall type, start/end times, and seat availability.

Purpose: Enables customers to select a preferred showtime.

System Requirements: Sessions must reference valid cinema halls and prevent invalid scheduling.

4. Session Seat Map and Availability

Description: Displays all seats for the hall of a selected session, marking each as reserved or available.

Purpose: Provides the “choose your seats” user interface similar to real cinema apps.

System Requirements: Combine seat data with reservation/ticket data to identify occupied seats.

5. Validate Seat Availability Before Booking

Description: Checks whether selected seats have already been reserved for a given session.

Purpose: Prevents double-booking and ensures consistent reservation behavior.

System Requirements: Analyze ticket and reservation records for conflicts before inserting new reservations.

6. Checking most Profitable Cinema Hall

Description: This operation calculates the total revenue generated by each cinema hall based on all completed reservations and their associated payments. By aggregating the final paid amounts for every session hosted in each hall, the query identifies which halls are financially the strongest performers.

Purpose: Helps management evaluate the profitability of each cinema hall, compare performance across hall types (e.g., VIP vs Regular).

System Requirements: Join cinema_hall, session, reservation, and payment tables; aggregate payment.final_amount by hall; ensure accurate financial tracking of all sessions hosted in each hall.

7. Cancel Reservation and Refund Payment

Description: Allows a customer to cancel a reservation, invalidates related tickets, and triggers refunds if applicable.

Purpose: Supports real-world booking flexibility and refund policy compliance.

System Requirements: Update reservation status, ticket validity, and generate a negative payment if required.

8. Customer Purchase and Ticket History

Description: Displays a list of all past reservations, associated tickets, seats, halls, and movies.

Purpose: Allows users to see their booking history, track past purchases, and manage loyalty rewards.

System Requirements: Join reservations, tickets, payments, sessions, halls, and movies.

9. Customer Genre Preference Profile

Description: Analyzes which movie genres each customer watches most frequently based on their past sessions and tickets. By counting reservations linked to movie genres, the system can determine a customer's preferred genres.

Purpose: Supports personalized recommendations and targeted marketing by helping the system understand each customer's viewing preferences.

System Requirements: Join reservation, session, movie to aggregate watched genres per customer, ensuring genre counts are based only on actual attendance.

10. Add New Movies and Assign Film Crew

Description: Returns all people who worked on a selected movie, including their role and biography.

Purpose: Helps admins or users view full crew information for a movie.

System Requirements: Uses the works_on relationship to join persons with the movies they contributed to.

11. Global Session Overlap Detection

Description: Generates a complete list of all overlapping sessions across the entire cinema, regardless of hall. It detects every conflicting pair of sessions.

Purpose: Gives administrators a full operational conflict audit, ensuring no scheduling mistakes exist anywhere in the cinema.

System Requirements: Perform a full conflict scan by self-joining sessions across all halls and detecting intersecting time ranges, supporting global schedule validation beyond EXCLUDE constraints.

12. Hall-Specific Session Overlap Check (During Scheduling)

Description: Checks if a newly scheduled session overlaps with existing sessions in the same hall. Used at the moment of creating or updating a session.

Purpose: Prevents double-booking a cinema hall.

System Requirements: Self-join sessions and use interval overlap logic.

13. Hall Utilization and Occupancy Analysis

Description: Calculates seat occupancy percentage for each session based on tickets sold.

Purpose: Helps management understand attendance patterns and optimize showtimes.

System Requirements: Aggregate tickets per session and compare with hall capacity.

14. Revenue Report by Date and Payment Method

Description: Summarizes revenue grouped by date and payment method.

Purpose: Provides financial analytics and helps management track the cinema's performance.

System Requirements: Summarize payment.final_amount by date and payment method.

15. Revenue per Movie

Description: Computes total revenue earned by each movie based on sold tickets.

Purpose: Shows which movies perform financially better to aid future scheduling decisions.

System Requirements: Join payment and reservation data through sessions and movies.

16. Top Customer Ranking

Description: Ranks customers by total money spent and number of confirmed reservations.

Purpose: Supports loyalty program management and targeted marketing.

System Requirements: Aggregate payments by customer.

17. Promotion Usage and Effectiveness

Description: Analyzes how often each promotion was applied and how much revenue resulted from it.

Purpose: Helps management evaluate promotional strategies.

System Requirements: Join promotion with related payment records.

18. Employee Workload Overview

Description: Shows how many sessions each employee monitors, grouped by date.

Purpose: Supports employee scheduling, workload balancing, and performance evaluation.

System Requirements: Join employees, monitors, and sessions.